

FUNCIONES RACIONALES:

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$$x^2 - 2x + 5 = 0 \quad \longleftrightarrow \quad x = \frac{2 \pm \sqrt{4 - 20}}{2}$$

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$$\int \frac{x - 5}{x^2 - 2x + 5} dx$$

$$\frac{x - 5}{x^2 - 2x + 5} \quad \leftarrow \text{fracción simple}$$

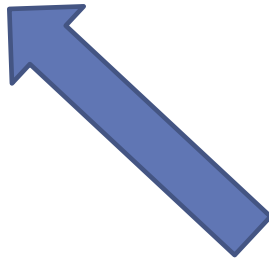
$$x^2 - 2x + 5 = 0 \quad \longleftrightarrow \quad x = \frac{2 \pm \sqrt{4 - 20}}{2}$$

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Primero buscaremos una integral inmediata de tipo logaritmo y después una integral inmediata de tipo arco tangente.

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$$\int \frac{x-5}{x^2-2x+5} dx = \frac{1}{2} \int \frac{2x-10}{x^2-2x+5} dx$$

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$$\int \frac{x-5}{x^2-2x+5} dx = \frac{1}{2} \int \frac{2x-10}{x^2-2x+5} dx = \frac{1}{2} \int \frac{2x-2-8}{x^2-2x+5} dx$$

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$$= \frac{1}{2} \int \frac{2x-2}{x^2-2x+5} dx - \frac{1}{2} \int \frac{8}{x^2-2x+5} dx$$

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$$\int \frac{1}{x^2-2x+5} dx = \int \frac{1}{(x-1)^2+4} dx$$

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$$\int \frac{1}{x^2-2x+5} dx = \int \frac{1}{(x-1)^2+4} dx = \frac{1}{4} \int \frac{1}{\left(\frac{x-1}{2}\right)^2+1} dx$$

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$$= \frac{1}{2} \int \frac{\frac{1}{2}}{\left(\frac{x-1}{2}\right)^2+1} dx = \frac{1}{2} \arctg \left(\frac{x-1}{2} \right) + C$$

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$$= \frac{1}{2} \ln |x^2-2x+5| - 2 \operatorname{arctg} \left(\frac{x-1}{2} \right) + C$$

$$\int \frac{1}{x^2-2x+5} dx = \int \frac{1}{(x-1)^2+4} dx = \frac{1}{4} \int \frac{1}{\left(\frac{x-1}{2}\right)^2+1} dx$$

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$$\begin{array}{r} 1 \quad -1 \quad -3 \quad -9 \\ 3) \quad \quad 3 \quad 6 \quad 9 \\ \hline 1 \quad 2 \quad 3 \quad 0 \end{array}$$

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no tiene raíces reales

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$$\int \frac{2x^2 + 7x - 3}{x^3 - x^2 - 3x - 9} dx$$

$$\frac{2x^2 + 7x - 3}{x^3 - x^2 - 3x - 9} = \frac{A}{x - 3} + \frac{Mx + N}{x^2 + 2x + 3}$$

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$$\frac{2x^2 + 7x - 3}{x^3 - x^2 - 3x - 9} = \frac{A}{x - 3} + \frac{Mx + N}{x^2 + 2x + 3} = \frac{A(x^2 + 2x + 3) + (Mx + N)(x - 3)}{(x - 3)(x^2 + 2x + 3)}$$

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$$2x^2 + 7x - 3 = A(x^2 + 2x + 3) + (Mx + N)(x - 3)$$

$x = 3$		$36 = 18A$		$A = 2$
$x = 0$		$-3 = 3A - 3N$		$M = 0$
$x = 1$		$6 = 6A - 2M - 2N$		$N = 3$

FUNCIONES RACIONALES:

$$\int \frac{2x^2 + 7x - 3}{x^3 - x^2 - 3x - 9} dx$$

$$\frac{2x^2 + 7x - 3}{x^3 - x^2 - 3x - 9} = \frac{2}{x - 3} + \frac{3}{x^2 + 2x + 3}$$

$$A = 2$$

$$M = 0$$

$$N = 3$$

FUNCIONES RACIONALES:

$$\int \frac{2x^2 + 7x - 3}{x^3 - x^2 - 3x - 9} dx = \int \left(\frac{2}{x - 3} + \frac{3}{x^2 + 2x + 3} \right) dx$$

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$$\int \frac{1}{x^2 + 2x + 3} dx = \int \frac{1}{(x + 1)^2 + 2} dx = \frac{1}{2} \int \frac{1}{\left(\frac{x+1}{\sqrt{2}}\right)^2 + 1} dx$$

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$$\begin{aligned} \int \frac{1}{x^2 + 2x + 3} dx &= \int \frac{1}{(x + 1)^2 + 2} dx = \frac{1}{2} \int \frac{1}{\left(\frac{x+1}{\sqrt{2}}\right)^2 + 1} dx \\ &= \frac{\sqrt{2}}{2} \int \frac{\frac{1}{\sqrt{2}}}{\left(\frac{x+1}{\sqrt{2}}\right)^2 + 1} dx \end{aligned}$$

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FUNCIONES RACIONALES:

$$\begin{aligned}\int \frac{2x^2 + 7x - 3}{x^3 - x^2 - 3x - 9} dx &= \int \left(\frac{2}{x - 3} + \frac{3}{x^2 + 2x + 3} \right) dx \\ &= 2 \ln |x - 3| + \frac{3}{\sqrt{2}} \operatorname{arctg} \left(\frac{x + 1}{\sqrt{2}} \right) + C\end{aligned}$$

$$\begin{aligned}\int \frac{1}{x^2 + 2x + 3} dx &= \int \frac{1}{(x + 1)^2 + 2} dx = \frac{1}{2} \int \frac{1}{\left(\frac{x+1}{\sqrt{2}} \right)^2 + 1} dx \\ &= \frac{\sqrt{2}}{2} \int \frac{\frac{1}{\sqrt{2}}}{\left(\frac{x+1}{\sqrt{2}} \right)^2 + 1} dx = \frac{1}{\sqrt{2}} \operatorname{arctg} \left(\frac{x + 1}{\sqrt{2}} \right) + C\end{aligned}$$

FUNCIONES RACIONALES EN $\sin x$, $\cos x$:

$$R(y, z) = \frac{y^2 z - z^2 + y - 1}{yz + y^2 - 2z + 3}$$

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$$R(y, z) = \frac{y^2 z - z^2 + y - 1}{yz + y^2 - 2z + 3}$$

$$R(\sin x, \cos x) = \frac{\sin^2 x \cos x - \cos^2 x + \sin x - 1}{\sin x \cos x + \sin^2 x - 2 \cos x + 3}$$



función racional en $\sin x$, $\cos x$

FUNCIONES RACIONALES EN $\sin x, \cos x$:

1. Si $R(-\sin x, \cos x) = -R(\sin x, \cos x)$ se realiza el cambio $t = \cos x$
2. Si $R(\sin x, -\cos x) = -R(\sin x, \cos x)$ se realiza el cambio $t = \sin x$
3. Si $R(-\sin x, -\cos x) = R(\sin x, \cos x)$ se realiza el cambio $t = \operatorname{tg} x$
4. Si no se verifica ninguna de las condiciones anteriores se realiza el cambio $t = \operatorname{tg} \left(\frac{x}{2} \right)$

FUNCIONES RACIONALES EN $\sin x, \cos x$:

Se sigue este orden

- 
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Se realiza el cambio $t = \cos x$

$$t = \cos x$$

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$$\arccos t = x$$



$$\frac{-1}{\sqrt{1-t^2}} dt = dx$$

$$t = \cos x$$



$$t^2 = \cos^2 x$$



$$t^2 = 1 - \sin^2 x$$



$$\arccos t = x$$



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$$\frac{-1}{\sqrt{1-t^2}} dt = dx$$



$$\operatorname{sen} x = \sqrt{1-t^2}$$

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$$\operatorname{sen} x = \sqrt{1-t^2}$$

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$$\operatorname{sen} x = \sqrt{1-t^2}$$

$$dx = \frac{-1}{\sqrt{1-t^2}} dt$$

$$\int \frac{\operatorname{sen}^3 x}{\cos x} dx = \int \frac{(1 - t^2)^{\frac{3}{2}}}{t} \left(\frac{-1}{\sqrt{1 - t^2}} \right) dt =$$


$$\cos x = t$$

$$\operatorname{sen} x = \sqrt{1 - t^2}$$

$$dx = \frac{-1}{\sqrt{1 - t^2}} dt$$

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$$= \int \left(t - \frac{1}{t} \right) dt = \frac{t^2}{2} - \ln |t| + C = \frac{\cos^2 x}{2} - \ln |\cos x| + C$$

$$\int \frac{\operatorname{sen} x \cos x}{1 + \operatorname{sen} x} dx$$

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$$R(\operatorname{sen} x, \cos x) = \frac{\operatorname{sen} x \cos x}{1 + \operatorname{sen} x}$$

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$$R(-\operatorname{sen} x, \cos x) = \frac{-\operatorname{sen} x \cos x}{1 - \operatorname{sen} x} \neq \frac{-\operatorname{sen} x \cos x}{1 + \operatorname{sen} x} = -R(\operatorname{sen} x, \cos x)$$

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Se realiza el cambio $t = \operatorname{sen} x$

$$t = \operatorname{sen} x$$

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$$\operatorname{arcsen} t = x$$



$$\frac{1}{\sqrt{1-t^2}} dt = dx$$

$$t = \operatorname{sen} x$$



$$t^2 = \operatorname{sen}^2 x$$



$$t^2 = 1 - \cos^2 x$$



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$$\cos x = \sqrt{1-t^2}$$

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$$dx = \frac{1}{\sqrt{1-t^2}} dt$$

$$\int \frac{\operatorname{sen} x \cos x}{1 + \operatorname{sen} x} dx = \int \frac{t\sqrt{1-t^2}}{t+1} \frac{1}{\sqrt{1-t^2}} dt$$


$$\operatorname{sen} x = t$$

$$\cos x = \sqrt{1-t^2}$$

$$dx = \frac{1}{\sqrt{1-t^2}} dt$$

$$\int \frac{\operatorname{sen} x \cos x}{1 + \operatorname{sen} x} dx = \int \frac{t\sqrt{1-t^2}}{t+1} \frac{1}{\sqrt{1-t^2}} dt = \int \frac{t}{t+1} dt$$

$$\begin{aligned}\int \frac{\operatorname{sen} x \cos x}{1 + \operatorname{sen} x} dx &= \int \frac{t\sqrt{1-t^2}}{t+1} \frac{1}{\sqrt{1-t^2}} dt = \int \frac{t}{t+1} dt \\ &= \int \left(1 - \frac{1}{t+1}\right) dt = t - \ln |t+1| + C\end{aligned}$$

$$\begin{aligned}\int \frac{\operatorname{sen} x \cos x}{1 + \operatorname{sen} x} dx &= \int \frac{t\sqrt{1-t^2}}{t+1} \frac{1}{\sqrt{1-t^2}} dt = \int \frac{t}{t+1} dt \\&= \int \left(1 - \frac{1}{t+1}\right) dt = t - \ln|t+1| + C \\&= \operatorname{sen} x - \ln(\operatorname{sen} x + 1) + C\end{aligned}$$

$$\int \frac{1}{1 - 2 \operatorname{sen} x \cos x} dx$$

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$$\int \frac{1}{1 - 2 \operatorname{sen} x \cos x} dx$$

$$R(\operatorname{sen} x, \cos x) = \frac{1}{1 - 2 \operatorname{sen} x \cos x}$$

$$R(-\operatorname{sen} x, \cos x) = \frac{1}{1 + 2 \operatorname{sen} x \cos x} \neq \frac{-1}{1 - 2 \operatorname{sen} x \cos x} = -R(\operatorname{sen} x, \cos x)$$

$$R(\operatorname{sen} x, -\cos x) = \frac{1}{1 + 2 \operatorname{sen} x \cos x} \neq \frac{-1}{1 - 2 \operatorname{sen} x \cos x} = -R(\operatorname{sen} x, \cos x)$$

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Se realiza el cambio $t = \operatorname{tg} x$

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$$\cos^2 x + \operatorname{sen}^2 x = 1 \quad \Rightarrow \quad 1 + \operatorname{tg}^2 x = \frac{1}{\cos^2 x} \quad \Rightarrow \quad \cos^2 x = \frac{1}{1 + \operatorname{tg}^2 x}$$

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$$\cos^2 x = \frac{1}{1+t^2}$$



$$\cos x = \frac{1}{\sqrt{1+t^2}}$$

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$$\operatorname{sen}^2 x = 1 - \frac{1}{1+t^2}$$



$$\cos^2 x = \frac{1}{1+t^2}$$



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$$\operatorname{sen}^2 x = 1 - \frac{1}{1+t^2}$$



$$\operatorname{sen} x = \frac{t}{\sqrt{1+t^2}}$$



$$\cos^2 x = \frac{1}{1+t^2}$$



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$$\cos^2 x = \frac{1}{1+t^2}$$



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$$\operatorname{sen} x = \frac{t}{\sqrt{1+t^2}}$$

$$dx = \frac{1}{1+t^2} dt$$

$$\int \frac{1}{1 - 2 \operatorname{sen} x \cos x} dx = \int \frac{1}{1 - 2 \frac{t}{\sqrt{1+t^2}} \frac{1}{\sqrt{1+t^2}}} \frac{1}{1+t^2} dt$$



$$\cos x = \frac{1}{\sqrt{1+t^2}}$$

$$\operatorname{sen} x = \frac{t}{\sqrt{1+t^2}}$$

$$dx = \frac{1}{1+t^2} dt$$

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$$= \int \frac{1}{\left(1 - \frac{2t}{1+t^2}\right) (1+t^2)} dt = \int \frac{1}{t^2 - 2t + 1} dt$$

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$$\int \frac{1}{1 + \operatorname{sen} x + \cos x} dx$$

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$$\int \frac{1}{1 + \operatorname{sen} x + \cos x} dx$$

$$R(\operatorname{sen} x, \cos x) = \frac{1}{1 + \operatorname{sen} x + \cos x}$$

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Se realiza el cambio $t = \operatorname{tg} \left(\frac{x}{2} \right)$

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$$\operatorname{sen} x = 2 \operatorname{sen} \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)$$

$$\operatorname{sen}(2a) = 2 \operatorname{sen} a \cos a$$

$$t = \operatorname{tg} \left(\frac{x}{2} \right) \Rightarrow \operatorname{arctg} t = \frac{x}{2} \Rightarrow 2 \operatorname{arctg} t = x \Rightarrow \frac{2}{1+t^2} dt = dx$$

$$\operatorname{sen} x = 2 \operatorname{sen} \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right) = \frac{2 \operatorname{sen} \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)}{\operatorname{sen}^2 \left(\frac{x}{2} \right) + \cos^2 \left(\frac{x}{2} \right)}$$

$$t = \operatorname{tg} \left(\frac{x}{2} \right) \Rightarrow \operatorname{arctg} t = \frac{x}{2} \Rightarrow 2 \operatorname{arctg} t = x \Rightarrow \frac{2}{1+t^2} dt = dx$$

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$$t = \operatorname{tg} \left(\frac{x}{2} \right) \Rightarrow \operatorname{arctg} t = \frac{x}{2} \Rightarrow 2 \operatorname{arctg} t = x \Rightarrow \frac{2}{1+t^2} dt = dx$$

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$$\cos x = \cos^2 \left(\frac{x}{2} \right) - \operatorname{sen}^2 \left(\frac{x}{2} \right)$$

$$\cos(2a) = \cos^2 a - \operatorname{sen}^2 a$$

$$t = \operatorname{tg} \left(\frac{x}{2} \right) \Rightarrow \operatorname{arctg} t = \frac{x}{2} \Rightarrow 2 \operatorname{arctg} t = x \Rightarrow \frac{2}{1+t^2} dt = dx$$

$$\operatorname{sen} x = 2 \operatorname{sen} \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right) = \frac{2 \operatorname{sen} \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)}{\operatorname{sen}^2 \left(\frac{x}{2} \right) + \cos^2 \left(\frac{x}{2} \right)} = \frac{2 \operatorname{tg} \left(\frac{x}{2} \right)}{\operatorname{tg}^2 \left(\frac{x}{2} \right) + 1} = \frac{2t}{1+t^2}$$

$$\cos x = \cos^2 \left(\frac{x}{2} \right) - \operatorname{sen}^2 \left(\frac{x}{2} \right) = \frac{\cos^2 \left(\frac{x}{2} \right) - \operatorname{sen}^2 \left(\frac{x}{2} \right)}{\operatorname{sen}^2 \left(\frac{x}{2} \right) + \cos^2 \left(\frac{x}{2} \right)}$$

$$t = \operatorname{tg} \left(\frac{x}{2} \right) \Rightarrow \operatorname{arctg} t = \frac{x}{2} \Rightarrow 2 \operatorname{arctg} t = x \Rightarrow \frac{2}{1+t^2} dt = dx$$

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$$\cos x = \cos^2 \left(\frac{x}{2} \right) - \operatorname{sen}^2 \left(\frac{x}{2} \right) = \frac{\cos^2 \left(\frac{x}{2} \right) - \operatorname{sen}^2 \left(\frac{x}{2} \right)}{\operatorname{sen}^2 \left(\frac{x}{2} \right) + \cos^2 \left(\frac{x}{2} \right)} = \frac{1 - \operatorname{tg}^2 \left(\frac{x}{2} \right)}{\operatorname{tg}^2 \left(\frac{x}{2} \right) + 1}$$

$$t = \operatorname{tg} \left(\frac{x}{2} \right) \Rightarrow \operatorname{arctg} t = \frac{x}{2} \Rightarrow 2 \operatorname{arctg} t = x \Rightarrow \frac{2}{1+t^2} dt = dx$$

$$\operatorname{sen} x = 2 \operatorname{sen} \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right) = \frac{2 \operatorname{sen} \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)}{\operatorname{sen}^2 \left(\frac{x}{2} \right) + \cos^2 \left(\frac{x}{2} \right)} = \frac{2 \operatorname{tg} \left(\frac{x}{2} \right)}{\operatorname{tg}^2 \left(\frac{x}{2} \right) + 1} = \frac{2t}{1+t^2}$$

$$\cos x = \cos^2 \left(\frac{x}{2} \right) - \operatorname{sen}^2 \left(\frac{x}{2} \right) = \frac{\cos^2 \left(\frac{x}{2} \right) - \operatorname{sen}^2 \left(\frac{x}{2} \right)}{\operatorname{sen}^2 \left(\frac{x}{2} \right) + \cos^2 \left(\frac{x}{2} \right)} = \frac{1 - \operatorname{tg}^2 \left(\frac{x}{2} \right)}{\operatorname{tg}^2 \left(\frac{x}{2} \right) + 1} = \frac{1 - t^2}{1 + t^2}$$

$$t = \operatorname{tg} \left(\frac{x}{2} \right) \Rightarrow \operatorname{arctg} t = \frac{x}{2} \Rightarrow 2 \operatorname{arctg} t = x \Rightarrow \frac{2}{1+t^2} dt = dx$$

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$$\cos x = \cos^2 \left(\frac{x}{2} \right) - \operatorname{sen}^2 \left(\frac{x}{2} \right) = \frac{\cos^2 \left(\frac{x}{2} \right) - \operatorname{sen}^2 \left(\frac{x}{2} \right)}{\operatorname{sen}^2 \left(\frac{x}{2} \right) + \cos^2 \left(\frac{x}{2} \right)} = \frac{1 - \operatorname{tg}^2 \left(\frac{x}{2} \right)}{\operatorname{tg}^2 \left(\frac{x}{2} \right) + 1} = \frac{1 - t^2}{1 + t^2}$$

$$\cos x = \frac{1 - t^2}{1 + t^2}$$

$$\operatorname{sen} x = \frac{2t}{1 + t^2}$$

$$dx = \frac{2}{1 + t^2} dt$$

$$\int \frac{1}{1 + \operatorname{sen} x + \cos x} dx = \int \frac{1}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \frac{2}{1+t^2} dt$$


$$\cos x = \frac{1 - t^2}{1 + t^2}$$

$$\operatorname{sen} x = \frac{2t}{1 + t^2}$$

$$dx = \frac{2}{1 + t^2} dt$$

$$\int \frac{1}{1 + \operatorname{sen} x + \cos x} dx = \int \frac{1}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \frac{2}{1+t^2} dt$$

$$= \int \frac{2}{1+t^2+2t+1-t^2} dt = \int \frac{1}{t+1} dt$$

$$\int \frac{1}{1 + \operatorname{sen} x + \cos x} dx = \int \frac{1}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \frac{2}{1+t^2} dt$$

$$= \int \frac{2}{1+t^2+2t+1-t^2} dt = \int \frac{1}{t+1} dt$$

$$= \ln |t+1| + C = \ln \left| \operatorname{tg} \left(\frac{x}{2} \right) + 1 \right| + C$$

$$\int \cos^2 x \, dx$$

$$\int \cos^2 x \, dx$$

No se siguen los pasos explicados anteriormente

$$\int \cos^2 x \, dx$$

$$1 = \cos^2 x + \sin^2 x$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

$$\int \cos^2 x \, dx$$

$$\begin{array}{r} 1 = \cos^2 x + \sin^2 x \\ + \\ \cos(2x) = \cos^2 x - \sin^2 x \\ \hline 1 + \cos(2x) = 2 \cos^2 x \end{array}$$

$$\int \cos^2 x \, dx$$

$$\begin{array}{r} 1 = \cos^2 x + \sin^2 x \\ + \\ \cos(2x) = \cos^2 x - \sin^2 x \\ \hline 1 + \cos(2x) = 2 \cos^2 x \end{array}$$



$$\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos(2x)$$

$$\int \cos^2 x \, dx = \int \left(\frac{1}{2} + \frac{1}{2} \cos(2x) \right) dx$$

$$\begin{array}{rcl} & 1 = \cos^2 x + \sin^2 x & \\ + & & \\ & \cos(2x) = \cos^2 x - \sin^2 x & \\ \hline 1 + \cos(2x) & = & 2 \cos^2 x \end{array}$$



$$\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos(2x)$$

$$\begin{aligned}\int \cos^2 x \, dx &= \int \left(\frac{1}{2} + \frac{1}{2} \cos(2x) \right) dx = \frac{x}{2} + \frac{1}{4} \sin(2x) + C \\ &= \frac{x}{2} + \frac{1}{2} \sin x \cos x + C\end{aligned}$$

$$\begin{array}{rcl} & 1 = \cos^2 x + \sin^2 x & \\ + & & \\ & \cos(2x) = \cos^2 x - \sin^2 x & \\ \hline 1 + \cos(2x) & = & 2 \cos^2 x \end{array}$$



$$\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos(2x)$$

$$\int \operatorname{sen}^2 x \, dx$$

$$\begin{aligned}\int \operatorname{sen}^2 x \, dx &= \int \left(\frac{1}{2} - \frac{1}{2} \cos(2x) \right) dx = \frac{x}{2} - \frac{1}{4} \operatorname{sen}(2x) + C \\ &= \frac{x}{2} - \frac{1}{2} \operatorname{sen} x \cos x + C\end{aligned}$$

$$\begin{array}{r} 1 = \cos^2 x + \operatorname{sen}^2 x \\ \hline \cos(2x) = \cos^2 x - \operatorname{sen}^2 x \\ \hline 1 - \cos(2x) = 2 \operatorname{sen}^2 x \end{array}$$



$$\operatorname{sen}^2 x = \frac{1}{2} - \frac{1}{2} \cos(2x)$$

$$\int \operatorname{sen}(mx) \cos(nx) \, dx$$

$$\int \operatorname{sen}(mx) \operatorname{sen}(nx) \, dx$$

$$\int \cos(mx) \cos(nx) \, dx$$

$$\int \boxed{\sin(mx) \cos(nx)} dx$$

$$\int \boxed{\sin(mx) \sin(nx)} dx$$

$$\int \boxed{\cos(mx) \cos(nx)} dx$$

Se procede a escribir las funciones de otra forma

Usaremos las igualdades siguientes:

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\int \operatorname{sen}(9x) \cos(4x) dx$$

$$\operatorname{sen}(9x + 4x) = \operatorname{sen}(9x) \cos(4x) + \cos(9x) \operatorname{sen}(4x)$$

$$\operatorname{sen}(9x - 4x) = \operatorname{sen}(9x) \cos(4x) - \cos(9x) \operatorname{sen}(4x)$$

$$\int \operatorname{sen}(9x) \cos(4x) dx$$

$$\operatorname{sen}(9x + 4x) = \operatorname{sen}(9x) \cos(4x) + \cos(9x) \operatorname{sen}(4x)$$

$$+$$

$$\operatorname{sen}(9x - 4x) = \operatorname{sen}(9x) \cos(4x) - \cos(9x) \operatorname{sen}(4x)$$

$$\operatorname{sen}(9x + 4x) + \operatorname{sen}(9x - 4x) = 2 \operatorname{sen}(9x) \cos(4x)$$

$$\int \operatorname{sen}(9x) \cos(4x) dx$$

$$\begin{aligned} \operatorname{sen}(9x + 4x) &= \operatorname{sen}(9x) \cos(4x) + \cos(9x) \operatorname{sen}(4x) \\ + \\ \operatorname{sen}(9x - 4x) &= \operatorname{sen}(9x) \cos(4x) - \cos(9x) \operatorname{sen}(4x) \end{aligned}$$

$$\operatorname{sen}(9x + 4x) + \operatorname{sen}(9x - 4x) = 2 \operatorname{sen}(9x) \cos(4x)$$



$$\frac{1}{2} \operatorname{sen}(13x) + \frac{1}{2} \operatorname{sen}(5x) = \operatorname{sen}(9x) \cos(4x)$$

$$\int \operatorname{sen}(9x) \cos(4x) dx = \int \left(\frac{1}{2} \operatorname{sen}(13x) + \frac{1}{2} \operatorname{sen}(5x) \right) dx$$

$$\begin{aligned} \operatorname{sen}(9x + 4x) &= \operatorname{sen}(9x) \cos(4x) + \cos(9x) \operatorname{sen}(4x) \\ + \\ \operatorname{sen}(9x - 4x) &= \operatorname{sen}(9x) \cos(4x) - \cos(9x) \operatorname{sen}(4x) \end{aligned}$$

$$\operatorname{sen}(9x + 4x) + \operatorname{sen}(9x - 4x) = 2 \operatorname{sen}(9x) \cos(4x)$$



$$\frac{1}{2} \operatorname{sen}(13x) + \frac{1}{2} \operatorname{sen}(5x) = \operatorname{sen}(9x) \cos(4x)$$

$$\int \operatorname{sen}(9x) \cos(4x) dx = \int \left(\frac{1}{2} \operatorname{sen}(13x) + \frac{1}{2} \operatorname{sen}(5x) \right) dx$$

$$= -\frac{1}{26} \cos(13x) - \frac{1}{10} \cos(5x) + C$$

$$\int \operatorname{sen}(9x) \operatorname{sen}(4x) dx$$

$$\cos(9x - 4x) = \cos(9x) \cos(4x) + \operatorname{sen}(9x) \operatorname{sen}(4x)$$

—

$$\cos(9x + 4x) = \cos(9x) \cos(4x) - \operatorname{sen}(9x) \operatorname{sen}(4x)$$

$$\cos(9x - 4x) - \cos(9x + 4x) = 2 \operatorname{sen}(9x) \operatorname{sen}(4x)$$



$$\frac{1}{2} \cos(5x) - \frac{1}{2} \cos(13x) = \operatorname{sen}(9x) \operatorname{sen}(4x)$$

$$\int \sin(9x) \sin(4x) dx = \int \left(\frac{1}{2} \cos(5x) - \frac{1}{2} \cos(13x) \right) dx$$

$$\cos(9x - 4x) = \cos(9x) \cos(4x) + \sin(9x) \sin(4x)$$

—

$$\cos(9x + 4x) = \cos(9x) \cos(4x) - \sin(9x) \sin(4x)$$

$$\cos(9x - 4x) - \cos(9x + 4x) = 2 \sin(9x) \sin(4x)$$



$$\frac{1}{2} \cos(5x) - \frac{1}{2} \cos(13x) = \sin(9x) \sin(4x)$$

$$\int \operatorname{sen}(9x) \operatorname{sen}(4x) dx = \int \left(\frac{1}{2} \cos(5x) - \frac{1}{2} \cos(13x) \right) dx$$

$$= \frac{1}{10} \operatorname{sen}(5x) - \frac{1}{26} \operatorname{sen}(13x) + C$$

$$\int \cos(9x) \cos(4x) dx$$

$$\cos(9x - 4x) = \cos(9x) \cos(4x) + \sin(9x) \sin(4x)$$

$$+ \cos(9x + 4x) = \cos(9x) \cos(4x) - \sin(9x) \sin(4x)$$

$$\cos(9x + 4x) + \cos(9x - 4x) = 2 \cos(9x) \cos(4x)$$



$$\frac{1}{2} \cos(13x) + \frac{1}{2} \cos(5x) = \cos(9x) \cos(4x)$$

$$\int \cos(9x) \cos(4x) dx = \int \left(\frac{1}{2} \cos(13x) + \frac{1}{2} \cos(5x) \right) dx$$

$$\cos(9x - 4x) = \cos(9x) \cos(4x) + \sin(9x) \sin(4x)$$

$$+ \cos(9x + 4x) = \cos(9x) \cos(4x) - \sin(9x) \sin(4x)$$

$$\cos(9x + 4x) + \cos(9x - 4x) = 2 \cos(9x) \cos(4x)$$



$$\frac{1}{2} \cos(13x) + \frac{1}{2} \cos(5x) = \cos(9x) \cos(4x)$$

$$\int \cos(9x) \cos(4x) dx = \int \left(\frac{1}{2} \cos(13x) + \frac{1}{2} \cos(5x) \right) dx$$

$$= \frac{1}{26} \operatorname{sen}(13x) + \frac{1}{10} \operatorname{sen}(5x) + C$$